

C++ Function Wrappers

Jan Wilczek (thinkcell)

Berlin, Sep 28, 2026

Who am I?



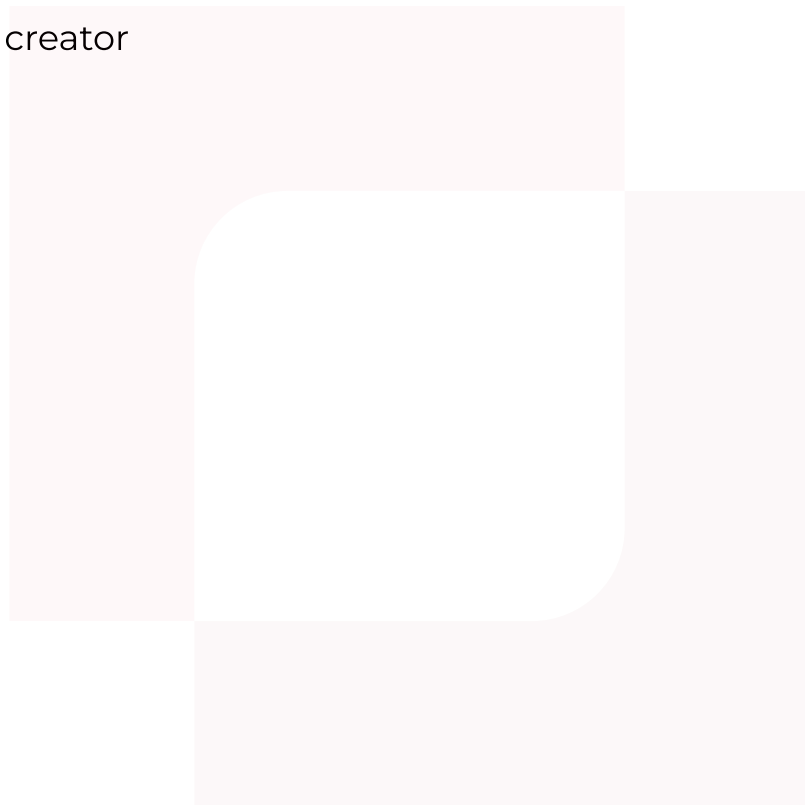
Who am I?

- Jan Wilczek [Yan Vil-check]



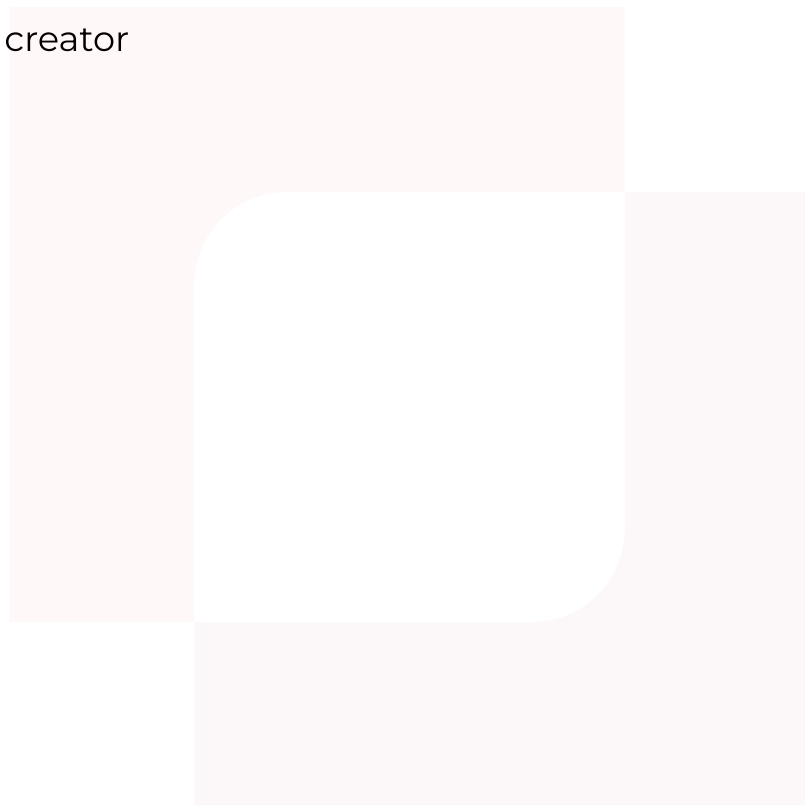
Who am I?

- Jan Wilczek [Yan Vil-check]
- Founder of TheWolfSound.com & online course creator



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- Jan Wilczek [Yan Vil-check]
- Founder of TheWolfSound.com & online course creator
- C++ Speaker at think-cell





Access 250+ best-practice templates

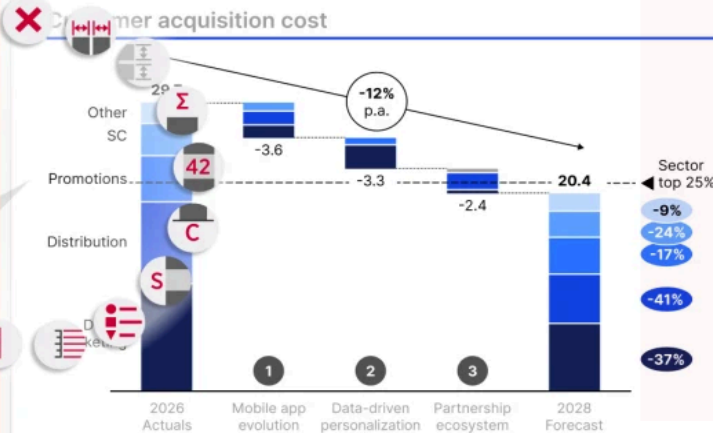


Use AI-powered tools inside PowerPoint and Excel

think-cell



Optimize customer acquisition cost by -12% p.a. to approach sector benchmarks by 2028



Key projects and funding required



Automate charts with Excel-linking

1. Mobile app evolution
 - Booking engine
 - Self-service purchase, seat selection
2. Data-driven personalization
3. Partnership ecosystem
4. 2028 Forecast

	A	B	C	D
1	Category	Digital mark..	Distribution..	Promotions..
2	2026 Actuals (Total)	13	10	5
3	Mobile app evolution	-3.6		
4	Data-driven personal..	-3.3		
5	Partnership ecosyst...	-2.4		
6	2028 Forecast	8	5	3



Create management-level charts and presentations



Fast & Furious



Fast & Furious



Fast & Furious



Fast & Furious



Fast & Furious



Fast & `std::function`



Fast & `std::function`

- `std::function`



Fast & `std::function`

- `std::function`
- `std::move_only_function` (C++23)



Fast & `std::function`

- `std::function`
- `std::move_only_function` (C++23)
- `std::function_ref` (C++26)



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- `std::function_ref` (C++26)
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Contents



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- Need for functions as first-class objects
 - const-correctness
 - copyability
 - movability
 - `noexcept`



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- Will `std::function` be deprecated?

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- Pictures of muscled men

Motivating Example: Logger

<https://godbolt.org/z/MGW3bv4rr>

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1 void doStuff() {  
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11 doStuff(logger);
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`std::function`



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- pass callables without explicit types or templates

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- pass callables without explicit types or templates
- functions as first-class objects
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- algorithms
- ABI stability

std::function

<https://godbolt.org/z/nG7ME146d>

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std::function: Problem 1 ("constness bug")

<https://godbolt.org/z/vKsd8cPe3>

```
1 auto f = std::function{[i = 0] mutable {  
2     ++i;  
3     std::println("i={}", i);  
4 }};  
5 const auto& fref = f;  
6 fref(); // i=1 ⚠
```

N4159

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std::function: Problem 1 ("constness bug")

<https://godbolt.org/z/vKsd8cPe3>

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1 auto f = std::move_only_function{[i = 0] mutable {  
2     ++i;  
3     std::println("i={}", i);  
4 }};  
5 const auto& fref = f;  
6 fref(); // ❌
```

N4159

"Awesome" logger

<https://godbolt.org/z/xK5bK7vx0>

```
1 namespace awe {
2     struct AwesomeLogger {
3         void log(std::string_view) & { /* ... */ }
4     };
5 }
6
7 void doStuff(std::function<void(std::string_view)>& log) {
8     // do stuff
9     log("Stuff done.");
10 }
11
12 auto logger = std::make_unique<awe::AwesomeLogger>();
13 doStuff(/* ? */);
```

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4     };
5 }
6
7 void doStuff(std::function<void(std::string_view)>& log) {
8     // do stuff
9     log("Stuff done.");
10 }
11
12 auto logger = std::function{
13     [logger = std::make_unique<awe::AwesomeLogger>()] (std::string_view str) {
14         logger->log(str);
15     }
16 };
17 doStuff(logger);
```

"Awesome" logger

<https://godbolt.org/z/xK5bK7vxo>

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4     };
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15     }
16 }; // ✗
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```

"Awesome" logger

<https://godbolt.org/z/xK5bK7vx0>

```
1 namespace awe {
2 struct AwesomeLogger {
3     void log(std::string_view) & { /* ... */ }
4 };
5 }
6
7 void doStuff(std::move_only_function<void(std::string_view)>& log) {
8     // do stuff
9     log("Stuff done.");
10 }
11
12 auto logger = std::move_only_function<void(std::string_view)>>{
13     [logger = std::make_unique<awe::AwesomeLogger>()] (std::string_view str) {
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```
std::move_only_function
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`std::move_only_function`



- C++23

`std::move_only_function`



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- non-copyable

`std::move_only_function`



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- allows move-only callables

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`std::move_only_function`



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`std::move_only_function`



- C++23
- non-copyable
- allows move-only callables
- allows copyable callables
- const-correct
- fixes a few more problems of `std::function` ...

std::function: Problem 2

<https://godbolt.org/z/WzsfjGrev>

```
1 auto f = std::function<void(void)>{[i = 0] {  
2     std::println("i={}", i);  
3 }};  
4 const auto& fref = f;  
5 fref(); // ✓
```

std::function: Problem 2

<https://godbolt.org/z/WzsfjGrev>

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2     std::println("i={}", i);  
3 }};  
4 const auto& fref = f;  
5 fref(); // ✗
```

std::function: Problem 2

<https://godbolt.org/z/WzsfjGrev>

```
1 auto f = std::move_only_function<void(void) const>{[i = 0] {  
2     std::println("i={}", i);  
3 }};  
4 const auto& fref = f;  
5 fref(); // ✓
```

std::function: Problem 3

<https://godbolt.org/z/P5EhvsoMv>

```
1 std::function<void(void)> get_f();  
2 // ...  
3 auto f = get_f();  
4 f(); // does this throw?
```

std::function: Problem 3

<https://godbolt.org/z/P5EhvsoMv>

```
1 std::move_only_function<void(void) noexcept> get_f();  
2 // ...  
3 auto f = get_f();  
4 f(); // if throws, program terminates
```

std::function: Problem 3

<https://godbolt.org/z/1zx8vKK6W>

```
1 auto f = std::move_only_function<void(void) const noexcept>{[] {  
2     std::println("I don't throw");  
3 }}; // ❌
```

std::function: Problem 3

<https://godbolt.org/z/1zx8vKK6W>

```
1 auto f = std::move_only_function<void(void) const noexcept>{[] noexcept {  
2     std::println("I don't throw");  
3 }}; // ✓
```

```
tc::move_only_function
```



Limitation of `std::move_only_function`

```
1 auto f = std::move_only_function<void(void) const noexcept> {[i = 0] noexcept {  
2     std::println("i={}", i);  
3 }};  
4 const auto g = f; // ❌
```

Limitation of `std::move_only_function`

```
1 auto f = std::copyable_function<void(void) const noexcept>{[i = 0] noexcept {  
2     std::println("i={}", i);  
3 }};  
4 const auto g = f; // ✓
```

`std::copyable_function`



↔ `std::move_only_function` +

`std::copyable_function`



↔ `std::move_only_function` +

- copy constructor

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- copy constructor
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↔ `std::move_only_function` +

- copy constructor
- copy assignment operator
- callables must be copy-constructible

Thread-safe logger

<https://godbolt.org/z/4YrE485sj>

```
1 void doStuff(std::move_only_function<void(std::string_view)>& log) {  
2     // do stuff  
3     log("Stuff done.");  
4 }  
5  
6 auto logger = [logger = std::make_unique<awe::AwesomeLogger>()] (std::string_view str) {  
7     logger->log(str);  
8 };  
9 doStuff(logger);
```

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1 void doStuff(std::function_ref<void(std::string_view)> log) {  
2     // do stuff  
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```
std::function_ref
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`std::function_ref`



- Non-owning callable wrapper

`std::function_ref`



- Non-owning callable wrapper
- Does for functions the same job as `std::string_view` for `std::string`

`std::function_ref`



- Non-owning callable wrapper
- Does for functions the same job as `std::string_view` for `std::string`
- Similar features to `std::move_only_function`
 - const correctness
 - `const` / `noexcept` / `ref` qualifiers
 - more

Composite logger

<https://godbolt.org/z/dxav63a7K>

```
1 struct CountingLogger {
2     CountingLogger(std::function<void(std::string_view)> logger)
3         : m_logger(std::move(logger)) {}
4
5     void operator()(std::string_view str) {
6         ++m_i;
7         m_logger(str);
8     }
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10 private:
11     std::function<void(std::string_view)> m_logger;
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10 private:
11     std::function_ref<void(std::string_view)> m_logger;
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13 };
14
15 auto logger = CountingLogger{[name = "foo"] (std::string_view str) {
16     std::println("{}: {}", name, str);
17 }};
18 doStuff(logger);
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```

```
tc::function_ref
```



Further differences from `std::function`

Ref-qualifiers

<https://godbolt.org/z/8KbMErcf>

```
1  auto f = std::function<void(void)>{ // cannot have & or && within <>
2      [] { std::println("Hello, Ref/Ref Ref/Val!"); } };
3  f();
4  std::move(f)();
5  std::function<void(void)>{
6      [] { std::println("Hello, Ref/Ref Ref/Val!"); } }();
```

Ref-qualifiers

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```

Ref-qualifiers

<https://godbolt.org/z/8KbMErcf>

```
1  std::move_only_function<void(void) &>{[] {  
2      std::println("Hello, Ref!");  
3  }}(); // ✖  
4  
5  auto f_ref = std::move_only_function<void(void) &>{  
6      [] { std::println("Hello, Ref!"); } };  
7  f_ref();  
8  
9  auto f_ref_ref = std::move_only_function<void(void) &&>{  
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11  f_ref_ref(); // ✖  
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3  }}(); // ✗
4
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```

No `target_type()` and `target()` accessors (requested by users and implementors)

<https://godbolt.org/z/6rxGvfGEW>

```
1 void foo() {
2     std::println("Hello, Type!");
3 }
4
5 auto f = std::function<void(void)>{foo};
6 std::println("{} ", f.target_type().name()); // prints "PFvvE"
7
8 if (void (*const* ptr)() = f.target<void(*)>()) {
9     std::println("Guessed it!");
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"Invocation has strong preconditions"

<https://godbolt.org/z/8b5fMbaoe>

```
1  std::function<void(void)>{}(); // std::bad_function_call
2  std::move_only_function<void(void)>{}(); // UB
3  std::copyable_function<void(void)>{}(); // UB
4  std::function_ref<void(void)>{}(); // ❌
```

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Why? 🤔



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https://www.reddit.com/r/cpp_questions/s/yxX2MXa4Yy

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- Implementations can now assert on this
- "Don't pay for what you don't use"
- Exceptionless environments
- `noexcept` propagation

```
1  std::throwing_move_only_function<void(void) noexcept> f;  
2  f(); // throws despite noexcept because empty  
3  f = [] noexcept { /* ... */ }; // assign
```

I guess Java/C#/Go wasn't getting enough backend projects and Rust wasn't getting enough low level work. Gotta take the opportunity to make C++ just a little more unsafe and make sure even the most up-to-date C++ still has a mountain of gotchas baked in. Don't worry; I'm sure there will be a safety profile for that later. (Yeah right.)

The language is too big to die, but not for lack of trying.

The C++ language development strategy at this is to pretty much fiddle while Rome burns basically.

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Conversions

From \ To	function	move_only_function	copyable_function	function_ref
function		✓	✓	✓
move_only_function	✗		✗	✓
copyable_function	✓	✓		✓
function_ref	✓	✓	✓	

Summary: `std::function` problems



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Summary: `std::function` problems

- can call a non-const callable via a const ref
- disallows move-only callables
- disallows non-movable, non-copyable callables
- does not propagate `const`, `noexcept`, `&`, or `&&` to `operator()`
- throws `std::bad_function_call`

Will `std::function` **be deprecated?**

Probably not 🙄

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But there is a proposal for it: P2721

Summary



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 - callables must be copy-constructible
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- never call an empty function wrapper (UB)